

STUDENT ACTIVITY

TensorFlow & Artificial Neural Networks — Activity

Activity: Architecture Design and Component Mapping

Q: Read the three model specifications below. Based on the project requirements, identify the appropriate **Activation Function** and **Loss Function** for each scenario.

Next, describe the purpose of an Optimizer during the training process in two to three sentences.

The Specifications:

1. **Model A:** A neural network designed for a fraud detection system that outputs a simple "Yes" or "No" to flag fraudulent transactions.
2. **Model B:** A computer vision network that classifies images of medical scans into multiple different disease categories.
3. **Model C:** A financial forecasting model predicting continuous stock market prices. Identify the specific activation function commonly used for its *hidden layers* and the loss function used to measure its continuous prediction errors.

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Architecture Design and Component Mapping

Model	Activation Function	Loss Function	Reason
Model A: Fraud detection system (Yes/No output)	Sigmoid	Binary Cross-Entropy	Sigmoid produces an output between 0 and 1 , making it suitable for binary classification . Binary Cross-Entropy measures the error between predicted and actual binary labels.
Model B: Medical scan image classification (multiple disease categories)	Softmax	Categorical Cross-Entropy	Softmax converts outputs into probabilities across multiple classes. Categorical Cross-Entropy evaluates the error for multi-class classification problems.
Model C: Stock market price prediction (continuous values)	ReLU (Hidden Layers)	Mean Squared Error (MSE)	ReLU is commonly used in hidden layers because it improves learning efficiency and reduces the vanishing gradient problem. MSE measures the difference between predicted and actual continuous values in regression tasks.

Purpose of an Optimizer

An **optimizer** is an algorithm that updates the neural network's **weights and biases** during training to minimize the loss function. It uses gradients calculated through **backpropagation** to move the model toward the optimal solution with the lowest prediction error. Common optimizers include **Gradient Descent**, **Adam**, **RMSProp**, and **SGD**, which help the model learn faster and more accurately.

Architecture Design and Component Mapping

Model	Activation Function	Loss Function
Model A – Fraud Detection (Binary Classification)	Sigmoid	Binary Cross-Entropy
Model B – Medical Scan Classification (Multi-Class Classification)	Softmax	Categorical Cross-Entropy
Model C – Stock Price Prediction (Regression)	ReLU (<i>Hidden Layers</i>)	Mean Squared Error (MSE)

Purpose of an Optimizer

An optimizer adjusts the **weights and biases** of a neural network during training to minimize the **loss function** and improve prediction accuracy. It uses gradients computed during **backpropagation** to update the model parameters efficiently. Optimizers such as **Adam**, **Gradient Descent**, and **RMSProp** help the model converge faster and achieve better performance.